BuildMind Complete Discussion Record

Project history decisions datasets and confirmed backup status

Prepared for the BuildMind project team and future project handoff. Updated 25 October 2026.

BuildMind is an AI assisted residential concept planning project for Sri Lanka. This record brings together product discussions, dataset research, architecture, prototypes, document reviews, recovery work and the completed Git backup steps.

Current outcome: the knowledge and recovered datasets are saved in the public knowledge repository. Five original source archives and seven cost PDFs were uploaded with Git LFS to a separate private data repository. Software development and source applicability review remain separate from file backup.

Document scope

This is a detailed consolidated conversation record rather than a verbatim message transcript. It preserves the available topics and technical decisions, with chronology and the latest confirmed status. A document record does not contain all external dataset bytes; repository locations and missing items are identified.

Record	Value
Edition	1.0 / 25 October 2026
Knowledge repository	https://github.com/Kogulanathan69/BuildMind-Knowledge-
Private data repository	https://github.com/Kogulanathan69/BuildMind-Data-Private
Latest knowledge commit evidenced	62f26ef
Latest private data commit evidenced	d33cc7f
Engineering or regulatory signoff	Not established by this conversation

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Parts 1 and 10 contain the final backup status. Earlier references to pending uploads are historical and are superseded by the confirmed commits stated here. Appendix A reproduces the earlier proposed BRD; its performance numbers remain targets and its original page references belong to the standalone BRD.

1 Latest status and essential context

Confirmed backup outcome

Repository	Saved content	Latest evidenced commit
BuildMind Knowledge	Discussion records, decisions, BRD, recovered datasets and documents	62f26ef
BuildMind Data Private	Five original ZIPs and seven cost PDFs using Git LFS	d33cc7f

The private repository push showed all seven new PDF LFS objects uploaded, main up to date with origin/main, a clean working tree and an empty git lfs status. Earlier commit 3bebda1 uploaded all five source ZIPs, totalling approximately 345 MB. Commit d33cc7f added seven cost PDFs, approximately 14 MB.

This completes backup of the recovered data/document/source batch selected in the conversation. It does not establish that every historical source, all missing images or the new handoff/document files have been committed. It also does not make the datasets production-ready.

User and delivery context

The user is a beginner developer using Windows and PowerShell, prefers Tamil with simple English, and wants this conversation kept to BuildMind. Seven members were planned for a two-month university project, with a preferred shorter completion period when practical. The user selected Member 3 room dimensions and space standards research, initially with two days for that research work.

Policies to preserve

- Sri Lankan statutory rules only. Do not treat foreign codes as local law.
- Global component dimensions are allowed; current Sri Lankan product availability and furniture prices remain future scope.
- Automatically generate candidate plans from the brief, rather than only displaying dummy templates.
- Support manual and conversational 2D edits, linked parametric 3D edits, automatic furnishing and subsequent customisation.
- RAG supplies evidence; deterministic geometry, solving, validation and measurement calculate the plan and quantities.
- No audited data-completion percentage, measured 99% engineering accuracy, permit or structural certification has been established.

Current implementation

The browser prototype demonstrates templates, dimensions, editing, illustrative objects, basic checks and Three.js views. Integrated source-backed RAG, constraint solver, regulatory applicability, climate rules and reviewed BOQ remain work to implement and test.

Remaining missing data

CubiCasa5K actual data, the nine standalone plan images, an older seven-field cleaned weather CSV, some cost-candidate CSVs and a completed Blender scene were not individually recovered. NBRO plan PDFs in the Niru collection do not establish recovery of those specific nine images.

2 Conversation chronology

1 Project concept

The user requested an AI construction design assistant, feasibility for Sri Lanka, global-market comparison, novelty, architecture, responsibilities and an achievable university delivery plan. BuildMind became the project name.

2 Engine list

The user listed geometry, constraints, circulation, openings, orientation, BOQ, cost, 2D, 3D and reporting. Layout generator/solver and scoring were identified as necessary additions.

3 Dataset sourcing

Repeated questions covered which data are needed, where to download them, how much had been collected, and how to turn the data into AI-readable Excel. No reliable overall completion percentage was established.

4 Team research ownership

Seven dataset areas were established. The user chose room dimensions/space standards as Member 3, with research/dataset preparation as the immediate responsibility. Adjacency source work was also discussed.

5 Statutory source policy

The user repeatedly required Sri Lankan rules only, excluding international codes such as ICC IRC from statutory research. Example plan dimensions and reviewed design recommendations must not be labelled law.

6 Drive and ZIP collection

The user asked how to upload PDFs to Drive, share them, inspect ZIPs, verify file counts and find missing sources. Some client screenshots were inaccessible, and download availability was repeatedly questioned.

7 Climate and components

Weather API connection was proposed. Raw observations must become reviewed design guidance. Global component dimensions were accepted; local supplier stock and current furniture product prices were moved to future scope.

8 Detailed product workflow

The user described brief-to-fit assessment, alternatives, editable 2D, AI comment editing, editable 3D, smart furnishing and user customisation. Shared geometry and revalidation were necessary consequences.

9 RAG and training

The user asked how RAG responds, whether to start, whether collected files train the model, and how to achieve accurate engineering-style plans. Evidence retrieval was separated from deterministic solving and calculations.

10 External plan datasets

CubiCasa5K and ResPlan were shared. Code clones were distinguished from data downloads, licences discussed and a Windows ResPlan clone failed due to lack of disk space.

11 Earlier experiments

n8n/SVG planning and Blender/MCP roof/furnishing experiments were discussed. They remain prototypes, not validated construction documentation.

12 UI revisions

The user rejected green and childish-looking interfaces and several redesigns. A Claude artifact/reference layout guided a simpler editor with a large model area and side panels.

13 Rendering improvements

The prototype moved to Three.js, locally vendored modules, hosted openings, numeric precision fixes and a fallback renderer. User screenshots still showed full-height walls hiding furniture and an overly long furnishing panel.

14 BRD creation

The user requested scope, USP, objectives, stakeholders, functional/nonfunctional requirements, integrations and stack. A formal BRD was generated, and a separate broad draft was reviewed for scope inconsistencies.

15 Submission edition

A 22-page BRD v1.1 dated 24 October was delivered as Word and PDF with bounded MVP, later scope, 40 FR, 15 NFR, engine ownership, evaluation and pending signoffs.

16 Knowledge repository

The user requested all discussions preserved in Git. Existing October 19 notes were retained and a comprehensive topic record plus BRD was committed as 3c4bb9b.

17 Bundle and authentication

No assistant GitHub authentication was available and the user declined the plugin. A bundle was provided; the user resolved file-location and account-permission errors and pushed successfully from Windows.

18 New PC discovery

The user clarified that this was a different PC without original documents. Earlier inventories had listed repository notes or another inventory, not the actual datasets.

19 Recovery

58 saved file versions, including 19 workbooks, were recovered. Raw weather was concatenated without filling gaps. Missing data and historical duplicate versions were documented.

20 Download problems

Two recovery ZIPs were created. Extraction reported empty ZIP and downloads failed, so smaller Excel/documents, weather/cost and component packs were supplied, followed by five individual source archive links.

21 Inventory confirmation

The user supplied an actual 118-file inventory showing an inner BuildMind-Data folder, datasets/documents, cost PDFs and five large source archives.

22 Recovered dataset push

The user copied datasets/documents into recovered-data, added limitations notes and pushed commit 62f26ef. A screenshot confirmed the new folder and commit on GitHub.

23 Private archive backup

Git LFS 3.7.1 was present on Windows. A private BuildMind-Data-Private repository was created; all five ZIPs uploaded successfully as commit 3bebda1.

24 Private cost PDF backup

Seven PDFs were copied, tracked with Git LFS and uploaded as commit d33cc7f. The user showed a clean working tree and no outstanding LFS objects.

25 Handoff and this record

A master context and file-transfer pack were generated for another chat/team member. The user then requested this discussion as one document, incorporating the final backup outcome.

3 Product scope and user journey

Identity and purpose

BuildMind is an AI assisted residential concept planning system for Sri Lanka. The product must generate a plan automatically, not just display a fixed dummy plan. The user wants dimensional consistency and realistic outputs; the requested “99% engineer accuracy” is an aspiration, not a measured or supportable claim. The system supports concept review, not automatic building permission or structural certification.

Intended complete workflow

1. Capture land width and length, units, locality and authority, north direction, road side, room requirements, parking, budget and preferences.

2. Confirm the structured brief, missing information, source applicability and assumptions.

3. Compute the usable envelope and use a constraint solver to search for candidate arrangements.

4. Explain candidate feasibility within the implemented model. If requirements conflict, identify the conflict and suggest changes for user approval rather than silently removing rooms.

5. Show distinct plan alternatives and explain their scores and tradeoffs.

6. Display a dimensioned 2D plan. Support manual dragging, numeric dimensions and supported natural language comments to change it.

7. Generate a parametric 3D view from the same model. Allow supported 3D edits because the user may discover new ideas while inspecting it.

8. After spatial concept review, place furniture, doors and bathroom fixture models intelligently; let the user move, replace, resize where appropriate and delete them.

9. Revalidate dependent geometry, access, openings and clearances. Recompute supported quantities and costs or mark them stale.

10. Save revisions, undo changes, finalise a concept and export a review report. Finalisation freezes a revision; subsequent edits create a new one.

Door openings, windows and bathroom fixture positions must be considered during planning. Waiting until furnishing to create essential access would invalidate earlier checks. Final furnishing selects models and appearance and refines placement.

Important user decisions

- Sri Lankan rules and regulations only for statutory research and validation. Do not substitute ICC IRC or another international code for Sri Lankan law.
- Global furniture, door and bathroom fixture dimensional sources are allowed. This exception concerns components, not regulatory requirements.
- Sri Lankan furniture stock, supplier availability and current product pricing are future scope. Sri Lankan construction BOQ/rates remain in scope as a bounded preliminary estimate.
- Consider road direction, north, daylight, ventilation and optional cultural preferences. “Pooja room should not be next to bathroom” is a user preference unless a separate applicable source proves a different classification.
- Keep this project record focused on BuildMind. Other projects discussed by the user are outside this repository.

Proposed first release and later expansion

The submission BRD proposes rectangular plots, one storey, one to three bedrooms, one or two bathrooms and optional one car parking. English first; Tamil may follow user testing. Include bounded 2D/3D edits, selected chat commands, a small verified generic component catalogue, one pilot jurisdiction, limited climate demonstration and a reviewed subset of BOQ and rates.

Irregular plots, additional floors, advanced roofs, broad authority coverage, unrestricted conversational editing, richer BIM exchange, uploaded survey interpretation and specialist simulation need later scope and estimates. Detailed structure, foundations, MEP design, geotechnical checks, procurement, payments and approval submission are not first release deliverables.

Stakeholders and roles

Homeowners provide facts and preferences; students and designers explore concepts; architects and planning reviewers review applicability and design guidance; civil/structural engineers advise within their discipline; QS reviewers validate measurement and rate interpretation. The project owner controls scope; the supervisor reviews academic acceptance. Data curators maintain evidence; admins manage authorised configuration and releases; developers and QA implement and test. A generic admin account is not a professional reviewer.

Proposed differentiators

Sri Lankan source-linked guidance; explainable fit and alternatives; one shared 2D/3D/quantity model; manual plus conversational editing; space-aware furnishing; explicit climate and preference tradeoffs. These are proposed USPs to demonstrate. Global novelty and competitor superiority have not been established. Maket, Homestyler and Drafted were mentioned for comparison; use the same benchmark briefs and report actual observations before claiming an advantage. Local design-service examples discussed include Structa and housedesign.lk, not validated feature baselines.

4 Architecture and AI responsibilities

RAG is an evidence service

The user asked whether to start RAG, how to train the collected data, how an agent would respond and whether plans could be generated directly. The agreed distinction is: source documents feed retrieval and reviewed rule extraction; geometry and layout generation require deterministic code and solving. Do not train a foundation model merely because PDFs have been collected. A useful pilot RAG can start with a small eligible corpus while further extraction continues, but its answers must abstain outside that coverage.

An agent response should state the understood requirements, ask for essential missing facts, cite applicable evidence, call the solver and validators, explain candidate results and propose reviewable edits. It must not produce invented room minima, rates or engineering quantities. An LLM cannot certify that a building can be constructed merely from land dimensions.

Twelve engines

Engine	Responsibility
Geometry	Exact coordinates, dimensions, polygons, shared walls, units and tolerances
Constraint	Plot envelope, applicable room bounds, hard constraints and soft preferences
Layout generator and solver	Search arrangements, preserve the brief and return truthful search status
Scoring	Rank eligible candidates using disclosed weighted soft criteria
Circulation validator	Entrance reachability, connected rooms, access paths and clearance
Opening validator	Wall hosting, offsets, door swings, window positions and source-linked conditions
Orientation	North and road context, reviewed light/ventilation/climate recommendations
BOQ	Reviewed measurement formulas, units, deductions and supported work items
Cost	Dated local rates, inclusions, exclusions and unpriced quantities
2D renderer	Dimensioned CAD-style plan and edit controls
3D renderer	Linked parametric model and supported edits
Validation report	Revision-specific results, evidence, pending checks and assumptions

The original list had ten engines. Layout generation and scoring were added because validators alone do not create plans. Assign implementation owners explicitly rather than assuming seven dataset owners cover all twelve engines.

Feasibility and scoring semantics

Keep candidate found, proven infeasible within the implemented constraint model, incomplete search/timeout, invalid model and insufficient information separate. A timeout does not prove impossibility. Even a feasible candidate does not establish soil suitability, utilities, structural safety or local approval. Missing statutory coverage remains pending.

Compare only candidates that meet implemented hard constraints. Record scoring criteria, weights, rule releases and score breakdown. Do not label an attractive rendering or arbitrary percentage as regulatory accuracy. Optional Vastu/cultural criteria are soft preferences by default.

Shared model and edits

Stable IDs for Project, Brief, Plot, Storey, Room, Wall, Opening, ComponentInstance, Rule, Evidence, ClimateContext, QuantityItem, RateItem, ValidationResult and Revision. Canonical units are integer millimetres in the proposed backend; render units are independent. OR Tools CP SAT requires an integer formulation; choose solver discretisation deliberately and validate resulting geometry.

A validated edit transaction changes the canonical model, rehosts openings, updates views and invalidates dependent validation/quantity/cost results. Preserve shared-wall relationships; dragging independent room rectangles is insufficient. Include owner checks, base revision, conflict handling, undo and safe recovery. An LLM emits allowlisted structured operations, not executable scripts. Source text cannot invoke tools or override system instructions.

Proposed implementation stack

Layer	Proposed baseline	Status
Frontend	Angular and TypeScript, CSS	Product direction; current prototype is plain JavaScript
2D	SVG/Canvas with geometry interaction; Konva considered	Final library not pinned
3D	Three.js and OrbitControls	Current prototype uses Three.js
API	FastAPI with Pydantic	Proposed backend
Geometry	Shapely	Proposed
Solver	OR Tools CP SAT	Proposed
Database and retrieval	PostgreSQL with pgvector	Submission BRD baseline
AI	One interchangeable LLM and embedding provider adapter	Provider and model unconfirmed
Documents	PDF extraction, optional OCR, evidence/page mapping	Pipeline not fully integrated
Files/jobs	File/object store and bounded background jobs	Proposed
QA	pytest and browser interaction tests, Git and CI	Acceptance work pending

An alternative uploaded BRD proposed PostgreSQL/PostGIS plus Chroma. Do not run both vector stores by accident. pgvector is the present submitted recommendation; PostGIS needs demonstrated spatial requirements before inclusion. Versions, production hosting and provider choices need confirmation. Blender/MCP is optional for assets/offline rendering; it is not necessary for every browser edit. n8n is an earlier orchestration experiment.

Integrations and API concepts

Candidate interfaces discussed: /api/projects, /api/generate-plan, /api/chat, /api/estimate-budget, /api/projects/{id}. These are design proposals, not evidence of deployed endpoints. Generation should receive project/revision IDs, confirmed brief and rule release, and return status, candidates, conflicts and evidence. Editing must use IDs, allowed operations and an expected base revision.

Weather integration supplies coordinates, period, timezone, temperature, rainfall, humidity and wind where available. Distinguish current weather, forecast and long-term climate. Provider failures preserve the plan and mark climate pending. Construction-rate ingestion is a curated import and review process, not an assumed live government API.

Capacity and spending

The user asked about 100 users/month and API limits. There is no fixed supported user count until model choice, tokens, calls per design, retries, concurrency, hosting and budget are measured. Use per-project budgets, caching and cancellation. No live OpenAI key configuration or provider entitlement is established by this knowledge record; do not commit keys.

5 Datasets sources and verification

Seven owners and eight collections

Member	Research area	Output
1	Weather and climate	Source weather plus reviewed climate rules
2	Construction cost	BOQ measurement, work items, resources and dated rates
3	Room dimensions and space standards	Sri Lankan sourced dimensional rules and reviewed examples
4	Room adjacency	Typed adjacency and access relationships
5	Building regulations	Applicable Sri Lankan planning requirements
6	Doors, windows, furniture and bathroom fixtures	Dimensions, clearances, swings, catalogues and rights
7	Architecture styles and templates	Structured plans, style metadata and evaluation examples

The user selected Member 3, research and dataset preparation, not ownership of automatic AI plan generation. Two days were available for that initial research task. Adjacency sources were also requested. Eight collections are planning rules, room standards, layout rules, climate rules, components, templates, BOQ/rates and test plans. Ten or twelve engines do not imply ten or twelve independent datasets; geometry, rendering and reporting are mostly code.

Common schema and governance

Record record_id, rule_type, severity, verification_status, verified_by, effective_date, unit, source document/URL, publisher, page/clause, jurisdiction, applicability conditions, collected date, version and notes. Keep evidence verification separate from local applicability. Types: regulation, recommended practice, architect guideline, prototype assumption, user preference. Severities: hard/soft. Draft values cannot silently become mandatory law.

Collection	Additional proposed fields
Room standards	room_type, parameter, min/max/recommended value, conditional basis
Layout rules	rule_id, room_a, room_b, relation, weight, rationale, reviewer
Climate rules	climate_zone, parameter, condition, recommended_value, basis, period
Components	component, type, width_mm, height_mm, depth_mm, thickness_mm, swing, clearance_mm, catalogue
Templates	template_id, plot range, bedrooms, storeys, plan_json, style, reviewer
Test plans	case_id, plot_w, plot_l, unit, brief_json, plan_json, expected_result, known_errors, reviewer, date

Collection	Additional proposed fields
BOQ/rates	work_item_id, unit, formula, deductions, resource coefficients, rate, location, date, currency, inclusions

Keep furniture in a separately understandable sheet where useful. Validate numeric ranges, units, required cells, IDs and references with pandas/Pydantic. A proposed starter completion gate was at least 20 source-verified sample rows, reviewer record, machine validation and Git versioning. This is not proof of sufficient coverage for a complete system. Never invent a source to fill a required cell.

Sri Lankan room and planning research

The user repeatedly excluded foreign codes including ICC IRC. UDA documents must be opened and checked for exact room provisions before a number is labelled a Sri Lankan legal minimum. If no applicable legal minimum is found, record that uncertainty; use a named architect-reviewed guideline where available. Example plan dimensions are examples, not statutory rules.

NHDA PDFs and some Plans.lk/Kedella Homes links were reported unavailable to download. University of Moratuwa housing research and Sri Lankan plan publications were requested as alternatives. A Scribd plan named 09-03 FIRST FLOOR-1 was questioned for Sri Lankan origin; origin and reuse rights must be checked, not inferred from its title. Four user-supplied PDFs needed separate extraction; an initial two-file count was disputed. This archive does not certify that those four PDFs are all physically present here.

Nine Sri Lankan floor-plan images were discussed. The user manually confirmed Plan 5 and Plan 8 are different. Treat all nine as distinct candidate plans unless evidence later shows duplication. Image-derived dimensions need scale and annotation verification. An NBRO pack previously discussed had 14 sheets, not necessarily 14 distinct houses, with separate long manuals/foundation references. Do not equate sheets, PDFs and unique layouts.

Existing deliverable references: BuildMind_NBRO_Reference_Plan_Extraction_v1.xlsx, BuildMind_AI_Room_Adjacency_Knowledgebase.xlsx, BuildMind_Room_Adjacency_DETAILED_FIXED.xlsx. The initial adjacency workbook was challenged as insufficiently populated, and a detailed corrected version was supplied. These filenames identify earlier deliverables; their workbook bytes and cells have not been newly audited by this update.

Weather

Existing repository report: Siyambalanduwa September-December 2021; 14,112 available ten-minute readings against 17,568 expected; 3,456 readings missing October 7-30. Timestamp coverage 80.3% is station-period coverage, not BuildMind completion. See knowledge/datasets/01_weather/DATA_QUALITY_REPORT.md. Solar, calibration, units, timezone and licensing need confirmation. One station and four months do not describe annual climate across Sri Lanka.

Open-Meteo was considered for API connection. Recorded reference https://open-meteo.com/en/pricing and documentation https://open-meteo.com/en/docs require current terms/period checks at integration. The prior review described free noncommercial evaluation/prototyping and paid commercial use. Do not promise a provider entitlement without checking it. Raw weather must become reviewed context-dependent guidance for opening direction, shading/overhang, ventilation pairing and any window-area recommendation. No universal numerical ratio or north-facing rule has been approved here.

Components

Global dimensional catalogues are allowed; verify each item's dimensions and use/installation clearances. Catalogue dimensions alone do not prove comfort, legal accessibility, safe swing or access. An earlier Member 6 pack reportedly

held 1,200 metadata entries with unverified dimensions and no per-row source URLs; 24 separate references were not matched to all entries. That count is not a verified catalogue. Sri Lankan product stock/prices stay future scope.

Plan datasets

ResPlan

Reference https://github.com/m-agour/ResPlan . A user upload ResPlan(1).zip previously passed ZIP integrity checks and contained ResPlan.pkl (258,453,658 uncompressed bytes; archive approximately 100 MB). This is prior discussion evidence, not a new record-by-record audit. The pickle was not unpickled. Never deserialize an untrusted pickle without an isolated review process.

Author-reported partition counts discussed: train 13,053; validation 1,632; test 1,632; augmented 683; total 17,000. Keep augmented/related plans together for evaluation and check near duplicates. Layout data are examples, not Sri Lankan permits. Plan graph conversion may be needed; no furniture/3D catalogue is implied. Prior review described data CC BY 4.0 and code MIT; retain the exact upstream terms and original attribution, and verify rights before redistribution.

The user's Windows clone failed with No space left on device and invalid index-pack output. Free storage and distinguish cloning code from acquiring dataset files; do not assert that a failed clone has installed the dataset.

CubiCasa5K

Reference https://github.com/CubiCasa/CubiCasa5k . Discussed as 5,000 plan images with 80+ object categories; actual dataset download was separate (around 5.5 GB in the prior review). Cloning the repository is not downloading its data. Prior review identified CC BY NC 4.0 restrictions; check terms and commercial suitability before adoption. It supports plan understanding, not local compliance or a ready furnishing model library. An earlier data.zip reportedly held code/model weights rather than the expected dataset.

Construction BOQ and rates

Existing cost notes remain in knowledge/datasets/02_construction_cost and the October 19 history. Sources discussed include Northern Province 2026 schedules and amendments, Department of Buildings BSR releases, provincial schedules, CIDA/ICTAD references, Census and Statistics bulletins and calculation references. Some collected documents are irrelevant, outdated or duplicates; a source archive inventory is not an approved rate catalogue.

Keep work items, measurement rules, resource coefficients, material prices, labour rates/productivity, equipment, transport, wastage and adjustment rules distinct. Preserve price history and district/effective-date selection. Separate RCC concrete, reinforcement, binding wire and formwork when the measurement basis requires it. Do not apply transport, tax, overhead or profit twice if included in a composite rate.

BSR/SoR are cost sources, not evidence for room minima or setbacks. Irrigation productivity is not automatically residential productivity. A prior schedule mentioned SLS 573:1999; a later review identified an SLS 573:2025 reference as a candidate update. The actual applicable measurement text/version remains unverified. Obtain the relevant document and QS review rather than claiming either edition has been adopted.

Unpriced quantities remain visibly unpriced; absence is not zero cost. Every estimate records revision, measurement basis, source release, date, region, currency, inclusions, exclusions and pending review. Preliminary quantities cannot imply structural design has been performed.

Source collection and completion

The user supplied a Google Drive folder and asked about uploading/downloading/sharing ZIPs, file counts, inaccessible links and remaining documents. The link is a collection location, not proof of access, completeness or rights. Missing

screenshot paths were repeatedly reported by the client; their contents cannot be reconstructed. The full private Drive link and personal email are intentionally not copied into this public-facing knowledge record.

Earlier approximate “70-80% source documents” and “25-35% cleaned” statements were not audited project metrics. Do not repeat them as current completion. Report separately: source acquired, extracted, source-verified, applicability-reviewed, schema-valid, integrated and acceptance-tested. Existing SOURCE_INVENTORY.csv, duplicate report and CSV catalogue are preserved; inspect their contents before counting.

Recorded source entry points

- UDA: https://www.uda.gov.lk/development-plans.html
- ResPlan: https://github.com/m-agour/ResPlan
- CubiCasa5K: https://github.com/CubiCasa/CubiCasa5k
- Open-Meteo: https://open-meteo.com/en/docs
- OR Tools: https://developers.google.com/optimization/cp/cp_solver
- pgvector: https://github.com/pgvector/pgvector

These are entry points, not direct download guarantees. Local rules require exact document, clause/page, applicable locality, date and reviewer.

6 Prototype experiments and UI feedback

Current prototype versus intended product

Prototype URL recorded in the project discussion: https://buildmind-design-studio.parthyniru27.chatgpt.site . Access is owner-restricted according to the last deployment record; the knowledge-repository update does not publish or change the site.

The current browser prototype demonstrates templated rooms, dimension editing, linked 2D and 3D views, generic furniture, basic boundary/overlap checks, local project state and keyword commands. It does not have integrated source-backed RAG, a real layout constraint solver, authoritative regulatory coverage, live climate rules or a validated BOQ/cost pipeline. Attractive output is not proof of those integrations.

Last recorded deployed source commit: b6e8627b921d9ef5fea0b6d29effe400ba936374 on 23 October. This is the separate prototype source repository, not a commit in the knowledge repository. The prototype source is not duplicated here by this documentation change.

UI evolution and user feedback

1. Initial green dashboard rejected; user requested a colour change.

2. Blue treatment and subsequent navy CAD/dashboard approaches were rejected as childish or unsuitable.

3. Visual home editor direction selected.

4. User shared a Claude artifact reference and screenshots with a large plan/model area on the left and requirement/edit cards on the right.

5. Reference-oriented redesign implemented; user complained that sizes/proportions and rendering were inconsistent.

6. Three.js rendering and fit/zoom were improved. Later Windows screenshots showed working 3D but full-height walls obscured interior furniture and the furnishing panel required excessive scrolling.

Desired UI: professional, simple, accurate proportions, large readable canvas, clear units, predictable dimensions and compact same-screen furnishing controls. Do not revert to a green playful dashboard. A real cutaway view and a compact furnishing library were identified as remaining work, not silently completed. A reference screenshot's arbitrary 65/100 score was not adopted as meaningful feasibility.

Browser implementation notes

The prototype uses plain JavaScript with viewer.js, locally vendored Three.js modules, OrbitControls and an official SVGRenderer fallback. WebGL was unavailable in the cloud QA browser; fallback rendering was inspected there. Later user screenshots showed actual Windows WebGL. Do not claim GPU testing in the cloud session.

Frontend state uses metres with feet display; proposed backend integer millimetres are a future model migration. Stable room/object/opening IDs and undo exist to a limited prototype extent. Local storage key was buildmind-visual-v1. Objects include beds, sofas, tables, chairs, plants, toilets, basins, doors and windows; they are illustrative, not a source-verified manufacturer catalogue.

Reference input: 40×60 ft, three bedrooms, two bathrooms, parking. Prototype assumed margin deductions produced a 34×35 ft footprint. These are demonstration assumptions, not Sri Lankan setbacks. Template zoning used bedrooms on the left, service rooms at the back and living/dining in the remaining area. Alternatives A/B/C were original and mirrored layouts, not independent solver-scored designs.

The revised renderer constructs hosted wall openings, sills/lintels and glass. Controls include room/object selection, move tools, roof visibility, wall-height slider and fit/reset. Numeric field precision was increased to avoid rounding

untouched dimensions into apparent overlaps. Mirror floating-point overlap tolerance was corrected. Shared-model 2D/3D transforms are bounded; full adjacency, legal clearances and BOQ invalidation are still not implemented.

SVG fallback draw order was corrected so floor/base/ground do not obscure interiors. PNG/JSON export paths were demonstrated. Full-height walls can still hide furniture; reducing wall height is not a substitute for a proper cutaway mode. Furnishing below the viewport remains a UX issue.

Earlier n8n and SVG experiment

Input fields: landWidth, landLength, bedrooms, bathrooms, parking, preferences. Earlier bounds included bedrooms 1-3, bathrooms 0-2, parking 0-1; the submitted proposed MVP now uses one or two bathrooms. Demonstration sizes included living 18×14 ft, kitchen 12×10, dining 10×10, master 12×12, other bedrooms 10×10, bathroom 6×8, parking 10×18 and corridor 4 ft. These were prototype assumptions, never verified Sri Lankan minima.

Rules tried: kitchen near dining, bedrooms grouped, master near living, north-light windows, open kitchen/dining merge. Output schema: {land:{width,length}, rooms:[], doors:[], windows:[]}; doors included coordinates, width, orientation and swing; windows included coordinates, width and orientation.

SVG experiment used a 900×760 canvas, auto scaling, land boundary, coloured rooms, labels, swing arcs, double-line windows, dimensions and north arrow, titled BuildMind AI Generated 2D Floor Plan with preliminary-concept wording. This was an experiment, not a tested CAD export engine.

Blender and MCP experiment

Prior user environment: Blender 3.6 with MCP on port 9876. Recorded Stage 4 roof Roof_Hip_22deg: 22-degree hip roof, 0.6 m overhang, 0.10 m solidify, terracotta Roof_Tile, collection 07_Roof. Camera BuildMind_IsoCam; EEVEE render 1920×1080, 64 samples, Filmic.

Stage 5 report described 64 objects: 3 land, 9 walls, 7 floors, 6 doors, 11 windows, 24 furniture, roof, camera, sun and original light. File name BuildMind_House_Stage5.blend; Stage 6 continuation was requested. That .blend file is not asserted present in this repository or freshly inspected. Do not turn a named scene snapshot into a validated construction model.

Branding

The user requested circular logo variants, merged visual references, lighter colours, removal of inner text and BuildMind name placement near/below the mark. Tagline direction: clear AI-assisted residential house design. Final logo bytes and a definitively approved variant are not established by this documentation update; do not invent a final asset.

7 Delivery budget and BRD history

University delivery

Seven-member project, two-month allowed window, preferred completion in roughly one to one-and-a-half months. Dataset research ownership does not imply ownership of all implementation engines. The user chose Member 3 room-dimensions research only for the immediate two-day research task. An eight-week integration plan is a proposal, not completed work.

Proposed sequence: week 1 scope/schema/pilot authority; week 2 reviewed starter data and persistence/retrieval; week 3 bounded solver/geometry; week 4 2D/openings/circulation; week 5 3D and structured edits; week 6 furnishing/climate/pilot BOQ; week 7 integration/exports/tests; week 8 evaluation and submission.

Budget

The user requested a total estimated cash budget of 15,000 for project API/AI work. The original request did not explicitly specify currency; the BRD carries LKR 15,000 as a confirmation-needed assumption. Proposed caps: inference 6,000; embeddings/ingestion 2,000; hosting/storage 4,000; review/tools 1,000; contingency 2,000. These are planning allocations, not current vendor quotes or sufficient professional consultancy fees. Earlier deliverable: BuildMind_Full_Cash_Budget_LKR_15000.pdf (not newly audited here).

Documents and diagrams requested

Simple Google-form description, Sri Lankan feasibility, global market and novelty comparison, timeline, requirements document, seven-member division, frontend/backend/AI split, system architecture, user journey and prototype were discussed/requested. A request does not prove a final deliverable exists. The formal BRD consolidates the latest bounded requirements; missing diagrams or files must be created or recovered explicitly.

BRD versions and review

1. An earlier complete BRD was generated as a 22-page DOCX/PDF, version 1.0. It covered scope, six proposed USPs, stakeholders, 40 functional requirements, 15 nonfunctional requirements, twelve engines, eight collections, twelve acceptance scenarios, proposed stack, integrations, eight-week plan, budget, risks, approval and references.

2. The user uploaded a separate 26-section text with 62 functional requirements and 20 nonfunctional requirements. It was reviewed as a broad product vision, not an accepted MVP baseline.

3. The user requested a submission BRD. Version 1.1, 24 October 2026, was delivered as BuildMind_Submission_BRD.docx and .pdf. Both are copied into docs/brd by this update. They contain proposed requirements, not completed test results or professional endorsements.

Review corrections for the 26-section draft: reconcile Must features with phase tables; bound the first-release geometry; explicitly own solver and scoring; distinguish timeout and infeasibility; make smart furnishing core; specify measurement/deductions/unpriced costs and QS review; map requirements to acceptance cases; choose a single vector-store baseline. Sections 5, 9, 12, 20 and 24 were identified for reconciliation. No automatic edits to that uploaded original text were claimed.

Submission details still require real team names, institutional details, supervisor details and pending approval/signature entries. Never fabricate approvals. Budget, pilot jurisdiction, reviewers, provider choice, test hardware and some release boundaries remain decisions for confirmation.

Evidence of readiness

Document formatting was rendered and visually reviewed as 22 pages. This says nothing about software acceptance. Version 1.1 records current prototype limitations and proposed targets. Follow the BRD's requirement IDs and acceptance scenarios when implementing; do not mix its 40 requirements with the alternative draft's 62 IDs without a deliberate mapping.

8 Open work and acceptance criteria

Priority backlog

1. Confirm pilot jurisdiction, project scope, named owners, reviewers and proposed spending envelope.

2. Audit source inventory, remove irrelevant and duplicate material, establish licences and applicability. Keep all nine user plans distinct unless evidence says otherwise.

3. Complete Member 3 source/page-based dimensional extraction. Label examples, guidelines and legislation correctly; do not import foreign statutory minima.

4. Produce typed adjacency rules, a verified small component catalogue and climate guidance; match each row to evidence.

5. Freeze canonical model and integer solver representation. Implement shared walls, opening hosts, room access, undo and revision contracts.

6. Build bounded layout solver and transparent scoring with truthful failure/search states.

7. Connect source-filtered RAG with page citations, abstention, prompt-injection resistance and structured operations.

8. Integrate manual and conversational 2D edits, linked 3D transforms and dependent revalidation. Fix cutaway and compact furnishing UX.

9. Implement space-aware automatic furnishing and replacement/move checks. Keep Sri Lankan product prices future scope.

10. Obtain QS-reviewed measurement rules; implement a small supported BOQ subset, dated rate releases and stale/unpriced handling.

11. Connect bounded climate demonstration with valid data periods and outage behaviour.

12. Run end-to-end tests, pilot usability evaluation and maintain a measured limitations report. Update Git knowledge after significant decisions.

Tests and proposed gates

Area	Cases to retain
Generation	Valid brief, oversized brief, missing input, forced timeout, invalid model
Geometry	Overlap, boundary escape, shared-wall resize, unit toggle within 1 mm
Circulation	Room without a door, unreachable room, blocked access path, door swing collision
Openings	Window outside wall, invalid host, missing bedroom window where an eligible rule requires one
Editing	2D/3D same revision, undo, conflicting edits, stale quantity/cost output
Furnishing	Footprint fit, clearance collision, replace/move and access preservation
RAG	Eligible citation/page, unavailable rule, conflicting sources, source injection
Cost	Known manual quantity example, deductions, composite-rate inclusions, missing price
Platform	Owner isolation, save/reopen/export, weather/API failure, rendering fallback

Test plans must contain intentionally invalid designs as well as valid ones. Expected results need reasons and a pinned rule release. The proposed starter set was 20 briefs, 20 invalid plans, 50 evidence-linked questions and 10 reviewed BOQ examples. Those are targets, not existing audited counts.

Proposed nonfunctional targets in the BRD include unit round trips within 1 mm, visible edits within 500 ms on the reference device, 30 fps for the bounded scene, configurable solver time budget initially 60 s and most RAG answers within 15 s excluding declared outages/ingestion. Measure on stated hardware; do not present targets as benchmarks already achieved. Proposed pilot: at least five consenting users if available, baseline comparison before claiming time savings.

Unknowns and access limitations

- The documentation is an organised record of available discussions, not a recoverable transcript of missing attachments or every historical message.
- Private Drive contents, missing client images and absent named spreadsheets are not certified present in this Git repository.
- Current local rule coverage, SLS measurement text, catalogue verification and dataset integration are incomplete.
- No reliable overall data-completion or engineering-accuracy percentage has been established.
- The full intended product exceeds a small initial release; proposed scope must be agreed rather than expanded silently.

Future handoff prompt

Read START-HERE.md, current status, project decisions and docs/discussion-record before work. Separate confirmed user policies, proposals, previous experiment behaviour and unverified claims. Keep Sri Lankan statutory sources only; allow globally sourced component dimensions; leave local product stock/pricing for future scope. Do not mistake the browser prototype for integrated RAG/solver/BOQ. Read the submission BRD and report actual test evidence before changing status.

9 Detailed topic coverage

Coverage is against the available project conversation/context and the repository read during this update, through 25 October 2026. This is a subject index, not a claim of complete historical message recovery. Recorded-but-missing files remain explicitly missing/unreviewed.

Discussed topic	Record
BuildMind name, residential purpose, Sri Lankan audience	PRODUCT-AND-SCOPE.md
Automatic generation rather than a dummy plan; engineering-quality aspiration	PRODUCT-AND-SCOPE.md
Land/brief/parking/budget/road/north inputs	PRODUCT-AND-SCOPE.md
Feasibility, explanations and alternative requirements	PRODUCT-AND-SCOPE.md; ARCHITECTURE-AND-AI.md
Multiple plans and transparent scoring	ARCHITECTURE-AND-AI.md
Manual and chat-driven 2D editing	PRODUCT-AND-SCOPE.md; PROTOTYPE-AND-EXPERIMENTS.md
Linked editable 3D and new ideas after viewing	PRODUCT-AND-SCOPE.md
Smart furniture, doors, bathroom fixtures and customisation	PRODUCT-AND-SCOPE.md
Openings/access planned before final furnishing	PRODUCT-AND-SCOPE.md
Geometry, shared model, revisions, undo and stale outputs	ARCHITECTURE-AND-AI.md
Original ten engines plus solver and scoring	ARCHITECTURE-AND-AI.md
RAG response, training confusion and deterministic modules	ARCHITECTURE-AND-AI.md
Starting a bounded RAG and continued data preparation	ARCHITECTURE-AND-AI.md
API/model cost, monthly limits and 100-user question	ARCHITECTURE-AND-AI.md; DELIVERY-AND-BRD.md
Sri Lankan statutory sources only; exclude international codes	PRODUCT-AND-SCOPE.md; DATASETS-AND-EVIDENCE.md
Global component dimensions allowed; local product prices future	PRODUCT-AND-SCOPE.md
Rule type, severity, verification, reviewer and applicability	DATASETS-AND-EVIDENCE.md
Seven team areas; Member 3 decision; two-day research task	DATASETS-AND-EVIDENCE.md
Eight collections and AI-readable Excel/CSV fields	DATASETS-AND-EVIDENCE.md
Climate API, raw weather versus reviewed design rules	DATASETS-AND-EVIDENCE.md
Four-month Siyambalanduwa data and missing period	DATASETS-AND-EVIDENCE.md; existing weather QA report
Room dimensions, adjacency and functional preferences	DATASETS-AND-EVIDENCE.md

Discussed topic	Record
Pooja/bathroom, north light and open-kitchen preferences	PRODUCT-AND-SCOPE.md; PROTOTYPE-AND-EXPERIMENTS.md
NHDA, Moratuwa, Plans.lk, Kedella and Scribd research issues	DATASETS-AND-EVIDENCE.md
Four-PDF count dispute and separate extractions	DATASETS-AND-EVIDENCE.md
Nine plans; Plans 5 and 8 distinct; NBRO sheet counts	DATASETS-AND-EVIDENCE.md
BSR/SoR, districts, amendments and irrelevant source filtering	DATASETS-AND-EVIDENCE.md; historical cost notes
Measurement standards, SLS uncertainty and QS review	DATASETS-AND-EVIDENCE.md
Work items, coefficients, RCC, units and rate analysis	DATASETS-AND-EVIDENCE.md; historical decisions
Admin updates, current rates and price history	DATASETS-AND-EVIDENCE.md; historical decisions
Wastage/transport/tax/overhead and double counting	DATASETS-AND-EVIDENCE.md
Component catalogue provenance/clearance shortcomings	DATASETS-AND-EVIDENCE.md
Templates, ResPlan and CubiCasa data/code distinction	DATASETS-AND-EVIDENCE.md
ResPlan ZIP integrity, split counts and untrusted pickle	DATASETS-AND-EVIDENCE.md
Dataset licences, attribution and noncommercial limits	DATASETS-AND-EVIDENCE.md
Windows no-space git clone failure	DATASETS-AND-EVIDENCE.md
Drive sharing, ZIPs, missing screenshots and download issues	DATASETS-AND-EVIDENCE.md
Data percentage questions and unaudited estimates	DATASETS-AND-EVIDENCE.md
n8n experiment, input limits, dimensions, JSON and SVG	PROTOTYPE-AND-EXPERIMENTS.md
Blender MCP, roof, Stage 5 scene and Stage 6 request	PROTOTYPE-AND-EXPERIMENTS.md
Logo variants and tagline requests	PROTOTYPE-AND-EXPERIMENTS.md
Green/childish UI rejection, Claude reference and redesign	PROTOTYPE-AND-EXPERIMENTS.md
Actual prototype versus proposed Angular/backend	ARCHITECTURE-AND-AI.md; PROTOTYPE-AND-EXPERIMENTS.md
Three.js, SVG fallback, wall openings, precision and mirrors	PROTOTYPE-AND-EXPERIMENTS.md
Remaining cutaway and long furnishing panel	PROTOTYPE-AND-EXPERIMENTS.md
Stack alternatives, PostGIS/Chroma versus pgvector	ARCHITECTURE-AND-AI.md
User/admin/reviewer responsibilities	PRODUCT-AND-SCOPE.md; BRD
USP, global-market/competitor comparison requests	PRODUCT-AND-SCOPE.md
Maket/Homestyler/Drafted benchmark suggestion	PRODUCT-AND-SCOPE.md

Discussed topic	Record
Architecture/user journey/requirements deliverable requests	DELIVERY-AND-BRD.md
Two-month delivery, shorter target and seven-member mapping	DELIVERY-AND-BRD.md
15,000 budget and LKR assumption	DELIVERY-AND-BRD.md
Original generated BRD, uploaded 26-section draft and review	DELIVERY-AND-BRD.md
Submission BRD, 40 FR, 15 NFR, scope/phases and approval	DELIVERY-AND-BRD.md; docs/brd
Invalid test plans, requirements mapping and evaluation	OPEN-WORK-AND-EVALUATION.md
Professional review, no permit/structural/99% claims	PRODUCT-AND-SCOPE.md; OPEN-WORK-AND-EVALUATION.md
Git continuity, handoff and preserving earlier notes	README.md; PROJECT-DECISIONS.md

Archive boundaries

Included bytes: repository's earlier knowledge notes/inventories, updated handoff records, discussion summaries, and the generated submission BRD in Word/PDF/Markdown. Not included by implication: private Drive archives, all original PDFs, every spreadsheet, nine plan images, ResPlan pickle, CubiCasa dataset, Blender project, complete prototype source or missing screenshots. Recover or link originals individually after rights/access checks; do not claim their bytes are backed up merely because this index mentions them.

10 Git recovery and backup operations

Locations confirmed in the conversation

Purpose	Windows location
Knowledge checkout	C:\Users\User\Downloads\BuildMind-Knowledge
Private data checkout	C:\Users\User\Downloads\BuildMind-Data-Private
Recovered originals	C:\Users\User\Desktop\BuildMind-Data\BuildMind-Data

Completed steps and evidence

Step	Recorded result
Knowledge discussion commit	3c4bb9b pushed to main
Recovered datasets and documents	62f26ef pushed to main
Original source archive LFS push	3bebda1; 5/5 objects, approximately 345 MB
Cost source PDF LFS push	d33cc7f; 7/7 objects, approximately 14 MB
Final ordinary Git status	main up to date with origin/main; working tree clean
Final LFS status	No objects pending push, commit or staging

Original archive register

Filename	Bytes
BSR-2021-2022(1).zip	71349467
BSR.zip	22152502
Build Mind Data Set - Niru.zip	108710775
Const_BSR_2020_final-compressed.zip	42923819
ResPlan(1).zip	100106537

Seven cost source PDFs

Filename	Bytes
BSR-2021_Construction.pdf	1704243
BSR-2023_Construction.pdf	590130

Filename	Bytes
BSR.pdf	1694154
BSR_2024_cons.pdf	3346427
Const_BSR_2020_final-compressed.pdf	1241259
SoR 2026 - Buildings.pdf	4969230
SoR 2026.pdf	866376

LFS configuration used

The private repository tracks source_archives/*.zip and sources/02_cost/*.pdf through .gitattributes. The large data objects and Git pointer commits were both uploaded. Do not remove LFS tracking or describe pointer files alone as all the source bytes.

Errors and fixes discussed

- Bundle clone from C:\Users\User failed because the bundle was in Downloads. Changing to the correct folder fixed it.
- Git commands outside a checkout returned not a git repository. The bundle had to be cloned before setting origin and pushing.
- A 403 identified cached credentials for another GitHub account. Browser authentication subsequently succeeded. Chrome login alone does not necessarily replace Git credentials.
- Creating an inventory under the checkout lists checkout files only. The actual dataset directory must be selected.
- An example Desktop path did not exist. This was a path error, not evidence that all originals had been lost.
- A CSV containing only another inventory file did not prove actual datasets were present.
- The inner BuildMind-Data folder was confirmed by the final inventory and used for copy commands.
- Windows reported an empty compressed folder and browser downloads failed. Smaller independent downloads and individual original archives were provided.
- LF to CRLF warnings during staging were nonfatal line-ending notices.
- A diff pager stopping at a colon can be exited with Q; it is not a failed Git operation.

Backup boundaries

The recovered batch selected for publishing is saved across the two repositories. Keep local copies as an additional recovery copy. The missing-data register remains applicable. Backup completion does not imply legal eligibility, dataset quality, accurate predictions or a finished software product. The new handoff files and this document are not asserted committed by the earlier commits.

Appendix A Submission BRD full text

This appendix reproduces the proposed BRD v1.1, dated 24 October 2026. It is retained as a requirements baseline. The internal document map refers to pages in the standalone 22-page BRD, not pages in this discussion record. Backup status in Parts 1 and 10 supersedes older file-collection status.

BuildMind Business Requirements Document

AI assisted residential planning for Sri Lanka

This document defines the business scope, users, product behaviour, data requirements and delivery criteria for BuildMind. It describes the intended system from a residential brief through feasibility screening, editable plans, furnishing and preliminary quantities and costs.

The proposed university release is a controlled residential concept design workflow. Its outputs support discussion and professional review. They do not establish building approval, structural safety or readiness for construction.

Document control

Field	Value
Document identifier	BM BRD 001
Version and date	1.1 / 24 October 2026
Status	Submission edition for academic review
Prepared for	BuildMind project team and university supervisor
Prepared by	BuildMind Project Team
Project team	Seven member project team
Delivery planning window	Eight weeks proposed within the stated two month project period
Approval status	No signatures or external professional endorsements recorded

How to use this document

Agree the release boundaries and open decisions before implementation. Convert the numbered requirements into implementation tasks and tests. Maintain each rule, plan, asset and rate with its provenance so the team can distinguish verified evidence from assumptions.

Planning baseline: the BuildMind project brief, product requirements and prototype review. Named technical references were checked on 24 October 2026. The source document collection has not been reaudited as part of preparing this BRD.

Document map and present status

Section	Subject	Page
1	Business problem and objectives	3

Section	Subject	Page
2	Release scope	4
3	Proposed unique selling points	5
4	Stakeholders and responsibilities	6
5	Business workflow	7
6	Requirements for project intake and generation	8
7	Requirements for editing and openings	9
8	Requirements for furnishing and climate	10
9	Requirements for knowledge costs and outputs	11
10	Decision rules and engine responsibilities	12
11	Data and knowledge governance	13
12	Nonfunctional requirements	14
13	Architecture and technical stack	15
14	Integrations and interfaces	16
15	User experience and business use cases	17
16	Acceptance and evaluation	18
17	Delivery plan and team ownership	19
18	Risks and budget	20
19	Assumptions decisions and approval	21
20	References and glossary	22

Implementation status

The current browser prototype demonstrates template layouts, dimension editing, 2D and 3D views, generic components, basic checks and keyword based commands. Source backed RAG, an integrated constraint solver, verified local applicability, live climate analysis and a validated BOQ and cost pipeline remain planned work. Existing prototype behaviour is not evidence that a requirement below has passed acceptance.

R1 means the proposed university MVP. R2 means an extension to the intended product after the MVP, subject to delivery capacity. Future scope means a separate later initiative. All performance and quality figures in this BRD are proposed targets, not measured results.

Business problem and objectives

Problem and opportunity

A homeowner can describe rooms and preferences without knowing whether they fit a plot, how access should work or what a change does to the rest of the design. Separate drawings, visualisations and estimates make early revisions difficult to follow. BuildMind will combine a structured brief, explainable checks and linked views in one concept planning workspace.

The first users are Sri Lankan residential clients, architecture students and professionals reviewing early concepts. The product should make design assumptions visible and preserve a clear handoff to an architect, engineer or quantity surveyor.

ID	Business objective	Proposed success measure
BO01	Identify whether a brief has a candidate layout within the supported model.	Every generation result distinguishes feasible candidate, proven infeasible within model, search incomplete and missing inputs.
BO02	Help users revise a concept without losing dimensional consistency.	All defined edit acceptance cases preserve stable IDs and matching 2D and 3D dimensions.
BO03	Provide traceable guidance about selected Sri Lankan requirements.	All supported regulatory statements in the acceptance set link to an eligible source and clause or page; unsupported requests abstain.
BO04	Reduce early concept iteration effort.	Pilot target: median completion time at least 20% below the same task in the agreed baseline workflow, with task quality held constant.
BO05	Expose quantities and preliminary cost assumptions.	Every estimated line shows units, measurement basis, rate source, location, date and exclusions.
BO06	Keep the university delivery manageable.	Deliver the agreed R1 vertical slice in eight weeks; obtain a recorded scope change before expanding it.

Business value and measurement

Record a baseline before claiming time savings. Pilot with at least five consenting target users if available, use identical briefs, retain task outcomes and report sample size. Do not present an unmeasured accuracy percentage, a dataset count or a successful visual rendering as proof of engineering correctness.

Release scope

R1 university MVP

Rectangular plots, one storey, one to three bedrooms, one or two bathrooms and optional one car parking. English interface first; explicit feet and metre display with a shared canonical model.

A structured project brief, location and authority context, road side, north direction, budget preference and a limited set of supported constraints.

Two or three distinct candidate layouts when the solver finds them, a transparent explanation when it cannot, and a comparison based on documented scoring weights.

Editable 2D plans and parametric 3D models, bounded room and object edits, selected natural language edit commands, undo and revision history.

Opening placement, room access checks and generic furniture and bathroom fixtures from a small verified catalogue; revalidation after changes.

A curated Sri Lankan knowledge collection with source citations, one selected pilot jurisdiction, a limited climate demonstration and a reviewed subset of BOQ and rates.

Save and reopen a project, export a concept report and plan data, show assumptions and distinguish pending checks from passed checks.

R2 intended product extensions

Extend the same workflow to more local authorities, broader component collections, richer conversational edits, more detailed preliminary BOQ items and more climate scenarios. Consider irregular plots, additional floors, complex roofs and richer exchange formats only after geometry, circulation and rule applicability are stable. R2 features require a new estimate and scope approval.

Future scope

Add Sri Lankan supplier product availability and current furniture, door and fixture prices; procurement and ordering; professional collaboration; automatic interpretation of uploaded plan images; richer BIM interchange; specialised accessibility workflows; energy simulation and optional immersive walkthroughs.

Explicit exclusions

R1 does not provide structural design, reinforcement or foundation calculations, geotechnical assessment, services engineering, approval submission, a certified cost quotation or construction documentation. Payments, supplier transactions, live collaboration and training a foundation model are outside R1. International building codes will not be substituted for Sri Lankan regulatory requirements.

Proposed unique selling points

The following are proposed differentiators to demonstrate and evaluate. This BRD does not claim that these features are globally unique or absent from competing products; a separate competitor study is required for those claims.

USP	Proposed distinction	User value and evidence
USP01	Sri Lankan evidence linked to a selected project context.	Explain which source and applicability conditions support a check; show uncertainty when local coverage is incomplete.

USP	Proposed distinction	User value and evidence
USP02	Requirement fit with understandable alternatives.	When a brief is too demanding for the supported model, show which constraints conflict and suggest changes without silently removing requirements.
USP03	One dimensioned model behind 2D, 3D, validation and quantities.	A room change updates every dependent view and flags stale outputs; demonstrate this with an end to end revision test.
USP04	Manual editing and conversational editing in the same workspace.	Users may drag an item or request an equivalent supported edit, with preview, undo and the same validation rules.
USP05	Space aware furnishing after concept review.	Place generic items using verified footprints, swing and clearance constraints, then let users replace and reposition them.
USP06	Explainable climate and preference tradeoffs.	Compare orientation, openings and optional cultural preferences using explicit soft criteria; do not represent a preference as a statutory rule.

Target users and needs

Homeowners need clear inputs, understandable plan options and safe revision controls. Students need traceable decisions and reproducible evaluation. Architects need editable geometry and a record of assumptions rather than an asserted approval. Quantity surveyors need a quantity basis and a dated rate set they can review.

Positioning

BuildMind is an AI assisted residential concept planning assistant for Sri Lanka. The initial business offering is a demonstrable design and explanation workflow. Commercial pricing, sales channels and service commitments remain undecided and are not implied by the prototype.

Stakeholders and responsibilities

Stakeholder	Interest and responsibility	Decision or access
Residential user	Provides plot facts and preferences; explores, edits and finalises a personal concept.	Own projects and exports; cannot change shared regulations.
Project owner	Maintains scope, priorities, delivery dependencies and the acceptance record.	Approves release scope and changes within the project.
University supervisor	Reviews academic objectives, feasibility and evaluation evidence.	Reviews academic acceptance; does not automatically certify construction suitability.
Architect or planning reviewer	Reviews selected room, circulation, opening and applicability rules.	Records domain review and identifies missing or conflicting evidence.
Civil or structural engineer	Identifies the boundary between concept and structural feasibility.	Provides advisory review; structural design remains excluded.
Quantity surveyor	Reviews measurement rules, item mapping and rate interpretation.	Approves the pilot measurement methodology and review status.
Dataset curator	Deduplicates sources and manages extraction, units, citations and versions.	Publishes candidate records for review; cannot self label uncertain values as law.

Stakeholder	Interest and responsibility	Decision or access
Administrator	Manages users, access, eligible knowledge releases and operational settings.	Administrative actions logged; does not edit private designs without authorisation.
Development and QA team	Implements the model, services, UI, integrations and evaluation.	Test environments and role appropriate project access.
Authorities and data providers	Supply rules, approvals processes, weather, rates or licensed assets.	External dependencies; no partnership or endorsement is assumed.

Role and ownership boundaries

R1 proposes owner, reviewer and administrator roles. Reviewers receive access only when a project is shared with them. Enforce ownership on every project and export request at the server. A domain review status is scoped to the reviewed record and version; it must not be displayed as general endorsement of BuildMind.

Nominate named people and confirm reviewer availability during week one. The BRD assigns responsibilities, not appointments or signatures.

Business workflow

Step	User and system action	Output or decision
1	User supplies plot dimensions, units, location, road side, north angle and desired spaces.	Validated brief with confirmed facts, preferences and missing inputs.
2	System selects eligible local rules and distinguishes hard constraints from soft preferences.	Rule context with sources and unresolved applicability.
3	Solver searches for candidate layouts and validators assess implemented constraints.	Candidate set, model scoped infeasibility, incomplete search or insufficient evidence.
4	User compares candidates, room areas and tradeoffs.	Selected initial concept; requirements are not silently relaxed.
5	User edits 2D manually or requests a supported conversational change.	A previewed revision, validation results and an undoable change.
6	The same model produces 3D; user inspects and makes supported parametric edits.	Updated 2D, 3D and affected checks for the same revision.
7	User confirms the spatial concept and requests furnishing.	Space aware furniture placement and fixture selections; doors and windows were already planned.
8	User moves, rotates, replaces or deletes items and reviews conflicts.	An adjusted furnished concept with pending issues visible.
9	System calculates supported quantities and preliminary costs.	Version matched BOQ subset, dated rates, exclusions and assumptions.
10	User finalises and exports the concept for professional review.	A frozen project revision and review package, not permission to build.

Iteration and recovery

Any edit returns the affected model to validation. Finalisation does not erase warnings. A changed model invalidates earlier quantities, costs and reports until they are recalculated. Failed API requests, cancelled searches and invalid proposed edits must preserve the last accepted revision.

Meaning of feasibility

A feasible result means a candidate satisfies the implemented model and its stated assumptions. It is not a statement about soil, structure, title, utility availability, hazards or local approval. Missing statutory evidence produces an unknown or pending check rather than an assumed pass.

Requirements for project intake and generation

Priority M means required for the agreed R1 release; S means a useful extension. Requirement acceptance must be demonstrated rather than inferred from a screen.

ID	Business requirement and expected behaviour	Verification	Release
FR01	M Create a named project with an owner, timestamp, unit setting and brief.	Save, reopen and retrieve only the authorised project.	R1
FR02	M Capture rectangular plot width and length, north angle, road side, locality and planning authority where known.	Reject invalid dimensions and identify missing authority context.	R1
FR03	M Record room counts, requested sizes, parking, budget and optional preferences.	Confirm a structured brief and mark facts versus preferences.	R1
FR04	M Convert feet and metres without changing the underlying geometry.	A unit toggle preserves dimensions within the stated tolerance.	R1
FR05	M Select an eligible rule release for the project context.	Inapplicable, draft or unresolved rules cannot establish a regulatory pass.	R1
FR06	M Calculate usable planning area using applicable rules or explicitly labelled assumptions.	Display the source and basis of each boundary deduction.	R1
FR07	M Generate up to three geometrically distinct candidate plans.	No overlap, required rooms present and all implemented hard checks satisfied.	R1
FR08	M Report candidate found, model infeasible, timeout or incomplete search, invalid model and missing information separately.	A timed out search is never labelled impossible construction.	R1
FR09	M Rank candidates using recorded soft criteria and weights.	Show a score breakdown; retain the same rule release for comparison.	R1
FR10	M Suggest requirement adjustments when conflicts are identified.	Explain the affected requirement; apply relaxation only after user agreement.	R1

The first jurisdiction and allowed room catalogue must be selected before FR05 and FR07 can pass. If statutory coverage is incomplete, a concept may still be explored using labelled assumptions, but its regulatory status remains pending.

Requirements for editing and openings

ID	Business requirement and expected behaviour	Verification	Release
FR11	M Render dimensioned 2D rooms, walls, openings, plot boundary, road side and north.	Labels and exported geometry refer to the same model revision.	R1
FR12	M Move and resize supported rooms using drag controls and numeric inputs.	Preserve stable IDs; recheck bounds, overlap, access and opening hosts.	R1
FR13	M Render parametric 3D from the shared model with orbit, zoom, roof and cutaway controls.	Changing the camera does not change room or object dimensions.	R1
FR14	M Support defined 3D edits to object transforms and room dimensions.	A 3D edit updates the source model and matching 2D view; no arbitrary mesh editing is implied.	R1
FR15	M Interpret selected conversational edit commands into structured operations.	Ask about ambiguous targets and preview changes affecting multiple spaces.	R1
FR16	M Record undo, redo and named project revisions.	Restore the exact prior model, selections and validation status as appropriate.	R1
FR17	M Host doors and windows on specific walls with offsets and sizes.	Reject openings outside their host or overlapping invalid wall segments.	R1
FR18	M Validate door swings, reachable room access and available circulation paths.	Detect disconnected rooms and obstruction cases using documented geometric criteria.	R1
FR19	M Check windows against the applicable rule context and room use.	Display missing or unsuitable openings with a rule reference or guideline label.	R1
FR20	M Revalidate affected geometry after each committed edit.	Block hard invalid commits or retain them only as an explicitly invalid draft; never mark them passed.	R1

Opening markers in the existing prototype do not satisfy FR17 by themselves. R1 requires actual wall hosting and coordinated updates. Movement and resizing must respect shared walls; moving a standalone rectangle is not sufficient for a connected house model.

Requirements for furnishing and climate

ID	Business requirement and expected behaviour	Verification	Release
FR21	M Maintain a curated global catalogue of furniture, doors and bathroom fixtures.	Each enabled item has source backed dimensions, unit, footprint and asset rights.	R1
FR22	M Place furniture according to room function and usable space after spatial concept review.	Keep required paths, door swings and approved clearances available.	R1
FR23	M Let users add, move, rotate, replace and delete catalogue items.	Use catalogue dimensions by default; flag custom resized products as conceptual.	R1
FR24	M Distinguish product size, use clearance and installation requirements.	Do not treat manufacturer installation data as Sri Lankan statutory rules.	R1

ID	Business requirement and expected behaviour	Verification	Release
FR25	M Capture location and north for a limited climate demonstration.	Show location, period, units and data source; allow manual climate context if the API fails.	R1
FR26	M Apply reviewed orientation and ventilation guidance as soft rules unless a sourced regulation applies.	Explain a recommendation and disclose model resolution and missing evidence.	R1
FR27	S Compare alternative opening and shading strategies across a wider climate set.	Use multi year or representative climate data; do not infer climate from today's forecast.	R2
FR28	M Store optional cultural and lifestyle preferences separately.	A preference such as separating a pooja room from a bathroom is user configurable and not labelled law.	R1
FR29	S Support richer fixtures, roof configurations and materials.	Each added parametric feature updates the model and relevant quantities through a documented mapping.	R2

Furnishing boundary

Generic global products are acceptable for concept design. Current Sri Lankan store availability, furniture prices, delivery charges and purchasing are future scope. The construction cost workflow may still use selected Sri Lankan building rates. No item should be costed twice as a building component and a separately priced furnishing.

Requirements for knowledge costs and outputs

ID	Business requirement and expected behaviour	Verification	Release
FR30	M Retrieve eligible passages from the curated knowledge collection.	Filter by document type, locality, date, rule status and access permission.	R1
FR31	M Answer supported questions with document, page or clause and source identity.	Abstain or qualify when evidence is absent, contradictory or outside scope.	R1
FR32	M Restrict AI edits to validated structured tool operations.	Document text cannot invoke tools, change system rules or execute code.	R1
FR33	M Allow curated document ingestion and reviewer publication.	Detect duplicates; retain extraction quality, source version and reviewer status.	R1
FR34	M Calculate a reviewed subset of BOQ quantities from the current model.	Show formula, measurement unit, deduction rule and measurement standard reference.	R1
FR35	M Map supported quantities to a selected Sri Lankan rate release.	Record district, year, revision, currency, taxes, transport and overhead basis.	R1
FR36	M Recalculate estimates after relevant model edits.	Stamp the revision; refuse to present an outdated estimate as current.	R1
FR37	M Produce an explainable validation and assumptions report.	Separate pass, fail, pending, not checked and not applicable by requirement.	R1
FR38	M Export concept drawings, PDF report and structured JSON.	Include units, scale information where applicable, version and construction limitations.	R1

ID	Business requirement and expected behaviour	Verification	Release
FR39	S Expand the BOQ catalogue and offer additional exchange formats.	Validate every new measurement mapping; preserve source provenance.	R2
FR40	M Finalise a revision and reopen it as a new editable revision.	Retain the original finalised record and its evidence snapshot.	R1

The pilot estimate excludes unmodelled structural, geotechnical, MEP and procurement work unless explicitly entered as a reviewed allowance. RAG can retrieve a rate explanation; deterministic measurement and costing services calculate the numbers.

Decision rules and engine responsibilities

Engine	Required responsibility
Geometry	Coordinates, dimensions, polygons, shared walls, units and tolerances.
Constraint	Applicable room bounds, plot envelope and configurable hard and soft constraints.
Layout generator and solver	Search feasible arrangements and return a truthful search status.
Scoring	Compare only eligible candidates using disclosed weighted preferences.
Circulation validator	Check room connectivity, entrance reachability and path clearance.
Opening validator	Check wall hosting, offsets, swings and applicable ventilation conditions.
Orientation	Combine north, road context and climate guidance without universal directional assumptions.
BOQ	Measure supported model elements using a reviewed measurement methodology.
Cost	Apply the selected dated rate release and show inclusions and exclusions.
2D renderer	Display dimensioned plan geometry and editing controls.
3D renderer	Display the same parametric model with supported edit operations.
Validation report	Aggregate evidence, assumptions and unresolved checks for the current revision.

Rule classification

Each rule records rule_type, severity, jurisdiction, applicability conditions, unit, effective period, verification status, source, page or clause and reviewer. Regulation, recommended practice, architect guideline, prototype assumption and user preference are different categories. Verified source text and verified local applicability are separate statuses.

Business decisions

Only applicable verified hard rules may support a regulatory pass. Missing mandatory evidence remains pending. A soft preference cannot override a hard rule.

Use one canonical model. A proposed edit is validated before acceptance; all derived views and quantities identify the resulting revision.

A failed search does not prove the physical site cannot be built on. Describe the supported model and any proved constraint conflict.

Finalising a concept freezes a revision; it does not issue a permit or certify engineering adequacy.

Data and knowledge governance

Collection	Minimum useful content	Publication condition
Planning rules	Authority, zone, setbacks, coverage and conditions.	Original source and pilot applicability reviewed.
Room standards	Room use, dimensions, area and conditional minima or recommendations.	No invented Sri Lankan legal minima; label guidelines correctly.
Layout rules	Adjacency, separation, access, severity and rationale.	Architect review; observation is distinct from prescription.
Climate rules	Climate context, parameter, period, condition and recommended action.	Reviewed transformation from weather to design guidance.
Components	Dimensions, footprints, swings, clearances, installation data and asset rights.	Verified catalogue values and licensed or original visual assets.
Templates	Geometry, room labels, connectivity, scale, provenance and licence.	Sample geometry audit and local revalidation.
BOQ and rates	Measurement method, item mapping and dated local rates.	QS reviewed pilot items; rate basis and exclusions recorded.
Test plans	Valid and deliberately invalid geometry with expected outcomes.	Independent holdout cases, review and version tracking.

RAG preparation and quality

Register sources, compute file hashes, remove duplicates and irrelevant material, extract text and tables, apply OCR where needed, then attach document and page metadata to chunks. Keep tables and conditional rules intact. Review critical numeric passages before indexing. Publish a versioned eligible collection and preserve it with project reports.

RAG retrieves supporting evidence; it does not train new foundation model weights. Structured rules and geometry remain accessible to deterministic services. Evaluate keyword and vector retrieval together, then assess answer support, citation accuracy and appropriate abstention.

Plan datasets and rights

ResPlan is a candidate source of vector plan examples, not a source of Sri Lankan regulations or approved site plans. Audit duplicates, scale, labels and connectivity before use; keep related plans in one evaluation split. CubiCasa5K supports plan image understanding research and is optional future work. Its noncommercial restriction must be respected [R7, R8]. The current BRD does not certify either dataset record by record.

Nonfunctional requirements

These are proposed R1 test targets. Record the selected hardware, browser, corpus and workload before measurement. Targets may be revised through the change process, not silently relabelled as achieved.

ID	Quality requirement	Acceptance target
NF01	Dimensional consistency	Canonical values retain at least millimetre precision; unit round trips differ by no more than 1 mm for supported sizes.
NF02	Interactive editing	On the agreed reference PC, 95% of tested committed local edits update visible geometry within 500 ms for the reference model.
NF03	3D responsiveness	Target at least 30 frames per second during orbit for a model of up to 15 rooms and 50 generic objects on agreed hardware; provide a 2D fallback.
NF04	Generation responsiveness	Use a configurable search budget, initially 60 seconds; display progress, cancellation and truthful timeout status.
NF05	RAG response time	Target 95% of test queries within 15 seconds, excluding clearly reported provider outages; log retrieval and generation latency separately.
NF06	Access control	All defined cross owner access tests fail closed for project, revision, document and export endpoints.
NF07	Data integrity	A failed save or invalid edit preserves the last accepted model. Use atomic revisions and explicit conflict handling.
NF08	Recovery	Proposed pilot recovery point is 24 hours and recovery time is 4 hours; demonstrate one backup restore before assessment.
NF09	Security	Server held API keys, TLS, input limits, rate limits, safe file parsing and sanitised display; no execution of uploaded pickle or document instructions.
NF10	Privacy	Collect only necessary plot and account data; restrict external AI context; support export and deletion with a documented retention policy.

ID	Quality requirement	Acceptance target
NF11	Accessibility and usability	Keyboard operable forms, visible focus, labelled controls, numeric alternatives to dragging and warnings that do not rely on colour alone.
NF12	Browser compatibility	Test the agreed Chrome and Edge releases on Windows; report WebGL fallback and unsupported devices explicitly.
NF13	Maintainability and testing	Versioned schemas, reviewed changes, geometry and validator tests, API tests and reproducible data releases.
NF14	Traceability and observability	Every check, estimate and AI edit records revision, rules, source release, operation status and errors without leaking secrets.
NF15	Pilot capacity	Exercise 10 simultaneous active sessions and 100 distinct demo projects; this is a test workload, not a monthly user or paid API entitlement.

Architecture and technical stack

Use a modular monolith for the first integrated release. The browser edits a shared project model; backend modules retrieve evidence, search layouts, validate changes and calculate outputs. Keep the layout solver separate from the LLM decision loop. Add distributed services only when measured workload or ownership requires them.

Layer	Proposed choice	Reason and boundary
Frontend	Angular with TypeScript and CSS	Forms, project workflow and typed state. Keep components simple for the team [R1].
2D editor	SVG or Canvas with a geometry interaction layer	Numeric editing, dimensions, selection, snapping and exports; enforce common model IDs.
3D editor	Three.js with parametric meshes and orbit controls	Model linked views, object transforms and ray based selection; 2D fallback for unsupported devices [R3].
Backend API	Python FastAPI with Pydantic	Typed contracts, input validation and integration with Python geometry and AI services [R2].
Geometry and search	Shapely and Google OR Tools CP SAT	Polygon checks and bounded constraint search. CP SAT requires integer constraints; scale to integer units explicitly [R4].
Database	PostgreSQL with pgvector	Projects, rules, revisions and vector retrieval in one database [R5].

Layer	Proposed choice	Reason and boundary
Document ingestion	PDF text extraction with optional OCR	Use page provenance and review queues. Pin parser versions after compatibility testing.
AI orchestration	One provider adapter for an LLM and embeddings	Choose provider and model after a cost and grounding evaluation; no provider is yet contracted.
Files and jobs	File object storage and a bounded worker queue	Source PDFs, exports and long running ingestion or generation jobs.
Testing and delivery	pytest, browser interaction tests and Git based CI	Reproducible checks, locked dependencies, separate configuration and a staged deployment.

Current prototype and migration

The existing prototype uses browser JavaScript and local project storage. It demonstrates interaction design, not the proposed secure multiuser backend. Retain useful model and renderer logic, add schema migration and replace browser only ownership with server authorisation. Blender may be an optional offline render tool; it is not required for every interactive edit.

Integrations and interfaces

Integration	Required exchange	Failure and operating rule
LLM provider	Filtered evidence and a user request; receive an answer or structured edit proposal.	Server credentials, timeouts, budgets and validation. No direct uncontrolled writes to projects.
Embedding service	Eligible document chunks and query embeddings.	Version the embedding model; rebuild an index when model compatibility changes.
Weather or climate API	Coordinates, period and variables; receive temperature, rain, wind and solar context.	Cache with provenance. On failure mark climate pending; do not invent observations.
Sri Lankan rule sources	Curated UDA, local plan and other eligible source documents.	Controlled ingestion rather than assuming a live authoritative regulations API.
Rates and measurement sources	Reviewed BSR or SoR releases and measurement references.	Manual or batch import with review; never treat a rate PDF as a source for room setbacks.
Global component sources	Manufacturer specifications and licensed 2D or 3D assets.	Track dimensions and rights; source sizes do not establish local availability or price.
Exports and storage	Versioned JSON, drawings, images and PDF reports.	Validate ownership, preserve units and revision, and expire temporary download links.

Weather licensing and design use

Open Meteo is a candidate provider. Its free tier is described for noncommercial evaluation and prototyping; commercial use requires a subscription. Historical data availability depends on the selected plan [R6]. Confirm the chosen endpoint and terms before integration. Multi year context is needed for orientation; a current forecast is not a climate design rule.

Core interface contract

A generation request contains project_id, base_revision, brief, unit and rule_release_id. Its result contains status, candidates, constraint_conflicts, pending_checks and evidence references. An edit request contains target IDs and a structured operation. The service returns a proposed or committed revision with validation results; it never silently overwrites a different base revision.

Stable entities include Project, Brief, Plot, Storey, Room, Wall, Opening, ComponentInstance, Rule, Evidence, ClimateContext, QuantityItem, RateItem, ValidationResult and Revision. Canonical dimensions use integer millimetres. Display units are independent; solver discretisation must not weaken verified bounds.

User experience and business use cases

Workspace requirements

Use the user supplied prototype references as the interaction starting point. Keep a large, proportionally correct 2D or 3D viewport and a readable requirements and review panel. Room names and dimensions must remain legible when furnished. Offer explicit mode, unit, north, road and revision indicators. Avoid hiding the active scene while the component library is open.

Provide a cutaway view to reveal interiors without changing real wall heights. Fit the full model to the viewport on first load and provide a reset camera control. Separate camera zoom from geometry scaling. Numeric fields must not round and resave untouched dimensions during unrelated edits.

Use case	Main path	Exception or recovery
UC01 Create a concept	Enter brief, confirm assumptions, generate, compare and choose a candidate.	Missing locality or rule coverage is shown before a regulatory claim can be made.
UC02 Adjust a room	Select a room, change size, preview consequences and commit a valid edit.	Show conflicts and permit undo; do not silently squeeze other rooms.
UC03 Edit through conversation	Ask for a supported change, resolve ambiguity, review the proposed operation.	Unsupported commands return a clear limitation and a manual edit route.
UC04 Inspect and furnish	Inspect 3D, confirm the spatial concept, place items and customise them.	Warn about clearance or access conflicts and offer alternate placement.
UC05 Review a cost	Select a rate release, calculate supported quantities and inspect exclusions.	Missing mappings are shown as unpriced; do not default missing costs to zero.
UC06 Finalise and export	Review warnings, freeze a revision and download the review package.	Show unresolved checks in the exported report; stale calculations must be recomputed.

Accessibility and language

R1 proposes English labels with plain explanations and unit clarity. Tamil and Sinhala support can follow user testing. Keep exact source excerpts in their original language and distinguish translated explanation from the authoritative source. Provide keyboard and numeric alternatives for drag actions.

Acceptance and evaluation

Release acceptance requires recorded evidence from the integrated R1 system. Screen appearance alone does not satisfy geometry, regulatory or cost requirements. Use a versioned holdout set and record both expected results and actual outputs.

Test	Acceptance scenario	Requirements
AT01	A supported brief produces candidates that contain the requested rooms and satisfy all implemented hard constraints.	FR01 to FR09
AT02	An oversized brief reports the solver status and conflicts. A forced timeout returns incomplete search, not construction impossibility.	FR08 FR10
AT03	Changing feet to metres and back preserves dimensions within 1 mm; display formatting does not change untouched fields.	FR04 NF01
AT04	A room resize updates shared walls, 2D, 3D and quantities; an invalid overlap cannot be finalised as passed.	FR11 to FR20
AT05	Undo restores the prior geometry, opening hosts and revision consistent outputs.	FR16 FR20
AT06	A disconnected room, blocked door swing and opening outside a host wall are correctly identified.	FR17 to FR19
AT07	Furniture replacement respects catalogue size; placement identifies tested access and clearance conflicts.	FR21 to FR24
AT08	A regulatory question receives an eligible source and page; absent or conflicting evidence produces abstention or pending status.	FR30 FR31
AT09	A malicious instruction inside a source document cannot invoke an edit or expose another project.	FR32 NF06 NF09
AT10	Known sample quantities match a QS reviewed worksheet within the stated item tolerance; rate basis is reproduced.	FR34 to FR36
AT11	Weather failure preserves the plan, labels climate pending and offers cached or manual context.	FR25 FR26
AT12	Saving, reopening, finalising and exporting preserve ownership, units, revision and outstanding issues.	FR01 FR38 FR40

Evaluation set and release gate

Proposed starter set: 20 diverse briefs, 20 deliberately invalid plan cases, 50 evidence linked questions and 10 reviewed BOQ examples. Keep related or duplicate source plans out of separate train and test partitions. All critical acceptance cases must pass; every remaining limitation must be listed in the release notes. Report measured citation support and abstention rather than claiming 99% overall engineering accuracy.

Delivery plan and team ownership

Week	Primary outcome	Evidence before progression
1	Freeze R1, pilot jurisdiction, canonical schema and benchmark briefs.	Scope record, initial source register, reviewer plan and schema examples.
2	Prepare verified starter rules and establish project persistence and knowledge retrieval.	Unit validation, source provenance and ownership tests.
3	Implement geometry, bounded generation and initial hard checks.	Valid and infeasible cases; truthful search status.
4	Connect editable 2D, openings and circulation checks.	Shared wall and door access tests; undo demonstration.
5	Connect parametric 3D and structured conversational edits.	Matching revision across views and ambiguity handling.
6	Add furnishing, limited climate context and pilot BOQ and rates.	Placement tests and reviewed quantity calculations.
7	Integrate reports, exports, performance checks and end to end scenarios.	Acceptance results, security tests and restored backup.
8	Run user evaluation, fix critical defects and package the submission.	Demonstration, limitations, source release and signoff record.

Seven member research and delivery mapping

Member	Research ownership	Suggested delivery pairing
1	Weather and climate rules	Orientation module and climate integration.
2	Construction measurement and rates	BOQ and cost service with QS review.
3	Room dimensions and space standards	Constraint rules, unit checks and applicability evidence.
4	Room adjacency and functional relationships	Solver preferences and circulation tests.
5	Sri Lankan planning rules	Rule governance and regulatory reporting.
6	Doors windows furniture and fixtures	Component catalogue and editable 2D or 3D assets.
7	Architecture templates and reference plans	Template retrieval, layout evaluation and UI workflow.

This is a proposed ownership model, not a claim that research members have accepted development tasks. The technical lead owns the solver integration; the adjacency lead owns scoring criteria. Confirm named assignees during kickoff. One member must coordinate integration and QA; domain data ownership alone does not deliver an automatic planning system.

Risks and budget

Risk	Business impact	Mitigation and owner
Incomplete local rule coverage	Users may mistake an assumption for a regulatory pass.	Pilot one locality, require provenance and retain pending status. Planning lead.
Solver complexity or failed search	Candidate generation may miss the delivery window.	Bound R1 geometry and search time; distinguish timeout. Solver owner.
Dataset errors or duplicates	Layouts and evaluation results may be misleading.	Audit sample records, units and near duplicates. Data curator.
2D and 3D drift	Users may edit or export inconsistent models.	Shared model, stable IDs and revision tests. Geometry owner.
LLM hallucination or unsafe instructions	Unsupported advice or unintended edits.	Evidence filters, abstention and validated operations. AI owner.
Unreviewed measurement or old rates	Preliminary costs may be materially wrong.	QS review, dated releases and explicit exclusions. Cost lead.
Device or external service limits	Rendering, weather or AI may be unavailable.	2D fallback, caching, cancellation and budgets. Platform lead.
Scope expansion or missing reviewers	The team cannot complete the agreed release.	Scope freeze, weekly review and early reviewer scheduling. Project owner.

Provisional cash planning envelope

A prior project planning figure of LKR 15,000 is carried forward as an assumption for confirmation. The following allocations are spending caps, not vendor quotes, exchange rate conversions or confirmed invoices. Team labour and existing equipment are not priced here.

Category	Proposed cap LKR	Basis
LLM and inference	6,000	Usage limited pilot only
Embeddings and ingestion	2,000	Prefer reusable indexes and local processing
Hosting and storage	4,000	Pilot environment and backups
Data review and tools	1,000	Small operational allowance, not a professional fee quote
Contingency	2,000	Provider, storage or integration variance
Total	15,000	Requires approval and current service quotes

Paid historical climate access, professional consultation, software licences and local product procurement are not assumed affordable within this cap. Use an eligible research workflow or revise scope and budget before committing expenditure.

Assumptions decisions and approval

Assumptions and dependencies

The university accepts a concept planning MVP and a transparent limitations report rather than an approved construction drawing service.

The team can provide an appropriate development PC, a persistent database and reviewer access; restricted university PCs may require a separate test environment.

Collected documents are candidate evidence until extraction, applicability, rights and reviewer status are established. No completion percentage is asserted by this BRD.

The team will confirm any statutory interpretation and measurement method with suitable reviewers. A downloaded source or dataset is not automatically applicable to a project.

Generic global component dimensions remain in scope. Current Sri Lankan furniture availability and prices remain future scope.

Open decisions before the R1 baseline

ID	Decision to confirm	Owner
D01	Select the pilot locality, planning authority and eligible rule release.	Planning lead and domain reviewer
D02	Confirm R1 limits, required room types and full versus reduced BOQ subset.	Project owner and supervisor
D03	Confirm the proposed technical stack and a compatible development environment.	Technical lead
D04	Select the LLM, embedding model, privacy settings and spending limits.	AI and project leads
D05	Choose the climate data period, provider access and reviewed guidance.	Climate lead
D06	Confirm KPI baselines, test hardware, measurement tolerances and pilot participants.	QA lead and reviewers
D07	Nominate reviewers and approve retention, sharing and finalisation policies.	Project owner

Change control and approval record

Record each requested change with its reason, affected requirement IDs, data dependencies, delivery impact and acceptance changes. The project owner and supervisor agree any change to R1; domain reviewers approve changes within their reviewed subject. Retain the previous BRD version.

Approval role	Name	Decision and date
Project representative	To be nominated	Pending
University supervisor	To be confirmed	Pending

Approval role	Name	Decision and date
Architect or planning reviewer	To be confirmed	Domain review pending
Quantity surveyor	To be confirmed	Measurement and rates review pending

References and glossary

References below support technology selection and source handling. They do not establish that a particular building rule applies to an individual site. Source documents used by the deployed RAG require their own document register with page, clause, effective period and applicability review.

R0 Project brief and scope baseline

BuildMind project requirements and prototype review, including automatic layout generation, manual and conversational edits, linked 2D and 3D, furnishing and future local product pricing.

R1 Angular documentation

https://angular.dev/overview

R2 FastAPI documentation

https://fastapi.tiangolo.com/

R3 Three js documentation

https://threejs.org/docs/

R4 Google OR Tools CP SAT documentation

https://developers.google.com/optimization/cp/cp_solver

R5 pgvector project documentation

https://github.com/pgvector/pgvector

R6 Open Meteo access and pricing

https://open-meteo.com/en/pricing

R7 ResPlan dataset and licence information

https://github.com/m-agour/ResPlan

R8 CubiCasa5K dataset and licence information

https://github.com/CubiCasa/CubiCasa5k

R9 UDA development plans and regulations source portal

https://www.uda.gov.lk/development-plans.html

Online technical references checked 24 October 2026. Pin implementation versions and recheck service terms at integration. The proposed stack is a project recommendation, not an already completed backend.

Glossary

Term	Meaning in this document
BRD	Business requirements document defining business intent and acceptance.
RAG	Retrieval augmented generation using selected evidence at answer time.
BOQ	Bill of quantities derived using a stated measurement method.
BSR and SoR	Building or schedule of rates; the precise release and scope must be recorded.
Hard constraint	A required condition that a candidate or committed edit may not violate.
Soft preference	A weighted design goal that can be traded off and explained.
Parametric model	Geometry defined by dimensions, relations and editable parameters.
Pending check	A requirement for which necessary data, applicability or validation is incomplete.

Appendix B Historical project decisions

The original decision register preserves earlier reasoning and cost design. Its October 19 current priorities and later pre-backup status are historical; they do not undo the completed uploads recorded in Part 10. Proposed choices still require the confirmation identified in the BRD.

BuildMind project decisions

Updated 25 October 2026.

Later decisions and precedence

This section takes precedence over conflicting October 19 wording below. Original decisions remain preserved; no old cost research is deleted.

ID	Decision or clarification	State
DECISION-033	Sri Lankan sources only for statutory rules; exclude ICC IRC and other foreign codes from regulatory research	User policy
DECISION-034	Global furniture/door/fixture dimensions allowed; current Sri Lankan product stock/prices are future scope	User policy; narrows product-pricing scope, not construction rates
DECISION-035	User takes Member 3 room-dimension research/dataset work, not immediate automatic-plan implementation	User task choice
DECISION-036	Manual and conversational 2D edits plus linked parametric 3D edits and smart furnishing are required product behaviour	User product intent
DECISION-037	Preserve shared geometry, revalidate edits, distinguish timeout from model infeasibility	Architecture requirement
DECISION-038	Add layout generator/solver and scoring to the original ten-engine list	Architecture correction
DECISION-039	Eight collections include climate rules, templates and deliberately invalid test plans	Data design
DECISION-040	One-storey rectangular bounded MVP and pgvector stack in submission BRD	Proposed baseline; scope/stack confirmation still needed
DECISION-041	No overall data-completion/engineering-accuracy percentage without an audited measure	Reporting rule
DECISION-042	All nine user plan images remain distinct; Plans 5 and 8 manually confirmed different	User verification
DECISION-043	Professional reference-oriented UI; green playful dashboard rejected	User design preference
DECISION-044	Submission BRD v1.1 records planned scope, pending approvals and explicit implementation limitations	Document baseline

ID	Decision or clarification	State
DECISION-045	Cost-only “current priority” in old DECISION-032 is historical; current work spans the integrated product	Supersedes current-priority wording, retains cost work

See docs/discussion-record for reasons, limits and open questions. Naming a role is not recording a person’s acceptance or professional signoff.

Historical October 19 decision register

The following text is retained as historical context. Apply the later scope clarifications above where relevant.

BuildMind Project Decisions

Last Updated: 2026-10-19 This file records important finalized BuildMind project decisions. Future developers, team members, or AI assistants should review this file before changing architecture, scope, datasets, or workflows.

DECISION 001 Project Name

Date: 2026-10-07 Decision: The project name is finalized as: BuildMind Status: Final Reason: The name represents an AI-assisted construction design and planning system.

DECISION 002 Main Project Type

Decision: BuildMind is an: AI-Powered Residential House Design and Construction Planning System Status: Final

DECISION 003 Primary Project Scope

Decision: The system should support:

- Residential house requirement collection
- Automatic floor-plan generation
- 2D visualization
- 3D visualization
- Cost estimation
- AI chat assistance
- Plan refinement
- Dataset-driven validation
- Admin-managed construction data

Status: Final

DECISION 004 Main System Roles

Decision: BuildMind has three major responsibility groups:

1\. User

2\. Admin

3\. AI

Status: Final

DECISION 005 AI Responsibility Boundary

Decision: The AI should understand user requests, explain outputs, and orchestrate system modules. The AI must not independently invent:

- Structural engineering values
- Construction quantities
- Construction prices
- Legal rules
- Safety certification
- Government approval status

Engineering and cost calculations must use deterministic formulas and validated data. Status: Final

DECISION 006 Frontend Technology

Decision: Angular will be used for the main web frontend. Status: Final

DECISION 007 Backend Direction

Decision: Python FastAPI is the preferred backend direction. Status: Current Final Direction Reason: It fits AI, optimization, geometry, and data-processing workflows well.

DECISION 008 Database

Decision: PostgreSQL is the preferred main database. Status: Current Final Direction

DECISION 009 3D Technology

Decision: Three.js is the preferred browser-based 3D visualization technology. Blender may be used for:

- prototyping
- rendering
- testing
- asset creation

Status: Final for MVP direction

DECISION 010 Floor Plan Generation

Decision: Floor-plan generation should not depend only on an LLM. The layout flow should use:

- structured user requirements
- room standards
- adjacency rules
- orientation rules
- climate rules
- constraint solving
- deterministic geometry
- validation
- scoring

The AI should support orchestration and explanation. Status: Final

DECISION 011 Multiple Plan Alternatives

Decision: The system should generate more than one valid floor-plan alternative when possible. The user should be able to compare alternatives. Status: Final

DECISION 012 Construction Cost Estimation

Decision: The AI must not guess the final construction cost. Cost estimation should follow: House Geometry → Measurement Engine → Work Item / BOQ Engine → Rate Analysis Dataset → Material Quantities → Labour Quantities → Equipment Quantities → Current Price Database → Location / District Adjustment → Wastage → Transport → Overhead → Profit → VAT / Tax → Final LKR Estimate Status: Final

DECISION 013 Dynamic Construction Prices

Decision: Construction prices must not be permanently hard-coded. The Admin Panel should allow authorized users to:

- Add prices
- Edit prices
- Set effective dates
- Update district-specific prices
- Activate prices
- Deactivate old prices
- Maintain price history
- Store source information

Status: Final Reason: Construction material and labour prices change over time and by location.

DECISION 014 Price History

Decision: Old price records should not simply be overwritten. Historical prices should be retained. Status: Final

DECISION 015 District Based Pricing

Decision: The cost system should support location-specific pricing. Priority regions for the MVP include Northern Province districts such as:

- Jaffna
- Kilinochchi
- Mullaitivu
- Mannar
- Vavuniya

Status: Final

DECISION 016 Current Price Selection

Decision: The cost engine should use the latest validated active price for the selected district or region. Status: Final

DECISION 017 Cost Source Priority

Decision: For Sri Lankan cost estimation, source priority should be:

1\. Northern Province Schedule of Rates

2\. Department of Buildings BSR

3\. CIDA / ICTAD references

4\. Department of Census and Statistics

5\. Provincial official schedules

6\. Verified supplier quotations

7\. Verified contractor quotations

8\. Civil-engineering reference books for calculation logic

Status: Final

DECISION 018 Dataset Traceability

Decision: Every important dataset row should eventually contain source metadata. Recommended metadata:

- source_name
- source_document
- source_url
- source_year
- source_region
- page_number
- collected_date
- verification_status
- notes

Status: Final

DECISION 019 Seven Member Dataset Split

Decision: Research responsibilities are divided into seven areas. Member 1: Weather / Climate Member 2: Construction Cost Member 3: Room Dimensions / Space Standards Member 4: Room Adjacency Member 5: Building Rules / Regulations Member 6: Doors / Windows / Furniture Dimensions Member 7: Architecture Styles / Plan Templates
Status: Final

DECISION 020 Cost Dataset Structure

Decision: The Construction Cost module should eventually contain structured datasets for:

- work items
- measurement rules
- resource coefficients
- material prices
- labour rates
- labour productivity
- reinforcement
- formwork
- masonry
- finishing
- plumbing
- electrical
- equipment
- transport
- adjustment rules
- source metadata

Status: Final

DECISION 021 Rate Analysis

Decision: Material prices alone are not enough for cost estimation. The system also needs resource-consumption data. Example: For 1 m^3 of a specified concrete work item, the system may need:

- cement
- sand
- aggregate
- water
- labour
- equipment

The exact coefficient must come from a validated source or reviewed engineering rule. Status: Final

DECISION 022 RCC Cost Structure

Decision: RCC cost should not be treated as concrete alone. RCC may include separate components such as:

- concrete
- reinforcement steel
- binding wire
- formwork
- concrete labour
- steel-fixing labour
- formwork labour

Status: Final

DECISION 023 Measurement Units

Decision: Work items should use appropriate engineering measurement units. Examples:

- Concrete → m^3
- Wall / Plaster / Tiles → m^2
- Steel → kg
- Doors / Fixtures → No.
- Pipes → m
- Labour → day / hour where applicable

Status: Final

DECISION 024 Admin Cost Module

Decision: The Admin Panel should eventually include a Cost Management module. Expected areas:

- Material Prices
- Labour Rates
- Work Items
- Rate Analysis
- District Rates
- Effective Dates
- Price History
- Data Sources

- Activate / Deactivate
- Import / Export

Status: Final

DECISION 025 Cost Estimate Disclaimer

Decision: BuildMind cost estimates should be presented as preliminary estimates. They are not:

- contractor quotations
- professional QS certification
- final structural quantities
- guaranteed construction costs

Status: Final

DECISION 026 Structural Validation

Decision: Structural quantities and safety-critical engineering values must be validated by qualified professionals before real construction use. Status: Final

DECISION 027 Building Rule Validation

Decision: BuildMind may assist with building-rule checks, but final legal approval must be verified with the relevant local authority or professional. Status: Final

DECISION 028 MVP Scope Control

Decision: The MVP should remain achievable within the university project timeline. MVP priority:

- basic residential planning
- automatic 2D plan generation
- room standards
- adjacency
- orientation
- basic 3D
- cost estimation
- AI interaction
- admin-managed datasets

Advanced BIM and full structural automation are not required for the MVP. Status: Final

DECISION 029 Knowledge Repository

Decision: BuildMind project knowledge should be maintained in a Git repository. Repository: BuildMind-Knowledge-Purpose:

- preserve project continuity
- support future ChatGPT sessions
- support team handoff
- preserve decisions
- preserve research
- preserve dataset definitions
- preserve next steps

Status: Final

DECISION 030 Handoff File Strategy

Decision: Future sessions should first read:

1\. START-HERE.md

2\. PROJECT-HANDOFF.md

3\. CURRENT-STATUS.md

4\. PROJECT-DECISIONS.md

5\. NEXT-STEPS.md

6\. README.md

Status: Final

DECISION 031 Repository Update Rule

Decision: After major BuildMind work: Update:

- CURRENT-STATUS.md
- NEXT-STEPS.md

If a new decision is finalized:

- PROJECT-DECISIONS.md

If architecture or scope changes:

- PROJECT-HANDOFF.md
- START-HERE.md

Then commit and push to Git. Status: Final

DECISION 032 Current Research Priority

Decision: The current main research priority is: Construction Cost Estimation Dataset Immediate focus:

- work items
- labour rates
- material prices
- rate analysis
- productivity norms
- reinforcement data
- adjustment rules
- source traceability

Status: Current Priority

Change Control Rule

Do not silently change a finalized decision. If a finalized decision must change:

1\. Add a new decision entry.

2\. Reference the old decision.

3\. Explain why it changed.

4\. Mark the old decision as superseded.

5\. Update relevant handoff documents.

Appendix C File and source registers

The following register lists recovered file versions with their original names. It is a file-preservation inventory, not a statement that all versions should enter one active RAG index. Retain one reviewed release and archive earlier or exact duplicate versions.

documents

Original recovered filename	Bytes
BuildMind_Full_Cash_Budget_LKR_15000.pdf	30083
BuildMind_Project_Requirements.docx	40642
BuildMind_Project_Requirements.pdf	142651
BuildMind_Submission_BRD.docx	69383
BuildMind_Submission_BRD.pdf	570654
BuildMind_University_Project_Cost_Estimation.pdf	79367

datasets

Original recovered filename	Bytes
BuildMind_25_District_AI_Dataset_Foundation.zip	4489
BuildMind_AI_Dataset_v1 (1)(1).xlsx	265365
BuildMind_AI_Dataset_v1 (1).xlsx	265365
BuildMind_AI_Dataset_v1 (1).zip	296183
BuildMind_AI_Dataset_v1.xlsx	265365
BuildMind_Architecture.xlsx	8151
BuildMind_M7_Dataset.docx	38798
BuildMind_M7_Dataset.zip	36502
BuildMind_Member6_Step10_FINAL_HANDOVER_ALL_IN_ONE (1).zip	14688521
BuildMind_Member7_Architecture_Dataset_Guide.docx	38798
BuildMind_Member7_Architecture_Dataset_Template.docx	39034
BuildMind_Member3_Final_4_Paper_Research_Dataset.xlsx	13046
BuildMind_Member3_Gazette_Verified_Dataset.xlsx	9081

Original recovered filename	Bytes
BuildMind_Member3_Research_Starter.xlsx	7627
BuildMind_Member3_Sri_Lanka_Room_Standards_Extract.xlsx	9827
BuildMind_Member3_Sri_Lanka_Room_Standards_Stage2.xlsx	14600
BuildMind_Member3_Stage3_Concept_Layout_Validation.xlsx	9054
BuildMind_Member3_Stage4_Verified_Research.xlsx	9677
BuildMind_Member3_Stage5_Sri_Lanka_Compliance_Research.xlsx	16400
BuildMind_Member3_Stage6_Gazette_Verification_Audit.xlsx	9684
BuildMind_Member3_Stage9_Gazette_Text_Verified.xlsx	9021
BuildMind_Member3_room_standards.csv	2256
BuildMind_NBRO_Reference_Plan_Extraction_v1.xlsx	10261
BuildMind_Stage8_2_Updated_Room_Dimensions.xlsx	10411
BuildMind_Stage8_Sri_Lankan_Floor_Plans_Initial_Extraction.xlsx	9467
ResPlan_demo.ipynb	152701
BuildMind_AI_Room_Adjacency_Knowledgebase.xlsx	49180
BuildMind_Room_Adjacency_DETAILED_FIXED.xlsx	43104
resplan_utils.py	29625
weather-data-siyambalanduwa.zip	3267670

experiments

Original recovered filename	Bytes
buildmind-3d-v2-architectural.zip	18851
buildmind-3d-v3-furnished.zip	17014
buildmind-angular-v2.zip	7880
buildmind-angular.zip	8527
buildmind-frontend-3d-roof-toggle.zip	15196
buildmind-frontend-3d.zip	10728
buildmind-frontend-sample.zip	11183
buildmind-frontend-selectable.zip	9579

Original recovered filename	Bytes
buildmind-frontend.zip	6530
buildmind-live-upgrade.zip	12792

sources

Original recovered filename	Bytes
BSR-2021_Construction.pdf	1704243
BSR-2023_Construction.pdf	590130
BSR.pdf	1694154
BSR_2024_cons.pdf	3346427
Const_BSR_2020_final-compressed.pdf	1241259
SoR 2026 - Buildings.pdf	4969230
SoR 2026.pdf	866376

source archives

Original recovered filename	Bytes
BSR-2021-2022(1).zip	71349467
BSR.zip	22152502
Build Mind Data Set - Niru.zip	108710775
Const_BSR_2020_final-compressed.zip	42923819
ResPlan(1).zip	100106537

Discussion and document sources

Project basis: the user requirements and messages available in this conversation; the topic records retained in the BuildMind knowledge repository; the submitted inventory; the generated BRD; and the user-supplied terminal outputs and screenshots confirming uploads. Original sources and their authority must be checked individually before they support design rules.

Technical entry points: Angular https://angular.dev/overview ; FastAPI https://fastapi.tiangolo.com/ ; Three.js https://threejs.org/docs/ ; OR Tools https://developers.google.com/optimization/cp/cp_solver ; pgvector https://github.com/pgvector/pgvector ; Open-Meteo https://open-meteo.com/en/docs ; UDA https://www.uda.gov.lk/development-plans.html . These are recorded reference links, not a new service, legal or licence audit.